

Deep Learning for Seizure Prediction Using Intracranial Electroencephalogram

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Abstract— One of the most prevalent and disabling neurologic disorders is epilepsy. Predicting epileptic seizures is a challenge to improve patients' quality of life, as approximately 30% of them do not respond to anti-epileptic drugs. Currently, EEG investigations are used in deep learning networks, which have recently gained significant success. The efficient utilization of these signals, particularly in disease prediction, is critical in terms of both time and cost. Based on that, we suggested a robust automated technique for classifying epileptic states in this paper, which was evaluated using an iEEG dataset collected from the American Epilepsy Society database. Our strategy is to build two deep learning models: CNN and CNN-SVM. The experimental results show that the proposed method based on CNN presents the best score which is 97.41% for accuracy, 97% for the F1 score, and 97.5% for the recall and precision.

Keywords—Epilepsy, seizure, prediction, EEG, deep learning, CNN.

I. INTRODUCTION

Epilepsy is one of the most prevalent neurological illnesses, affecting people of all ages, and it affects around 50 million people globally [1]. This pathology is a non-communicable chronic neurological disease. And it's a condition in which the central nervous system's electrical rhythms are disrupted. Anti-epileptic medications are used to regulate the activity of specific areas of the brain to prevent new epileptic seizures. Life, however, the medications are ineffective for roughly a quarter of the patients, and seizures persist [2].

An electroencephalogram (EEG) is a test that measures brain electrical activity. This non-invasive test is primarily used to diagnose epilepsy and assess the severity of an epileptic seizure. Occasionally, the superficial EEG does not provide all the information required for diagnosis and treatment selection.

As a result, EEG monitoring with implanted electrodes is sometimes required to provide additional information to the neurologist. This is known as intracranial electroencephalography (iEEG), and it is distinguished by the placement of electrodes beneath the surface of the skull. This method allows for a better understanding of cerebral activity over a longer period, allowing for a more accurate description of the disease. There are different major phases of seizures,

the EEG of an epileptic patient shows four different states: interictal is the period between seizures, preictal is the period before the seizure, ictal is the onset of the seizure, and finally postictal is the post-ictal phase.

Deep learning is an artificial intelligence subset evolved from machine learning, in which the system can learn on its own. This latter has recently attracted a lot of attention in a neurological illness diagnosis. It is used to improve health care by supporting patients and doctors in making a clinical diagnosis and developing a treatment plan more efficiently, speedily, and informed. In this regard, our contribution to this research is the construction of a system that permits the automatic prediction of crises. The first stage in this work is to transform the signal from its native domain to a frequency domain representation, the second step is to use classification techniques and the derived features to train a predictive model by differentiating between the periods of preictal and interictal of the iEEG dataset.

The remainder of this paper is structured as follows: Section II will introduce related work, Section III will present materials and methods, Section IV will present the results and discussion, and Section V will conclude the research.

II. RELATED WORK

The pre-seizure condition can be used to predict seizures. It starts a few minutes before the seizure and finishes with a seizure state. Previous works have extensively studied seizure prediction using iEEG. Many researchers focus on traditional machine learning methods, with the majority relying on Deep Learning-based algorithms that have been used to predict epileptic seizures.

The authors in [3] used STFT to generate spectrograms from the iEEG time series. They then fed them into a nine-layer CNN model to identify changes in the interictal and preictal iEEG signals. This model had a 75.00 % sensitivity to seizure prediction.

The authors of [4] suggested a prediction system based on a deep learning classifier that was taught to discriminate between signals. Its performance was evaluated using iEEG data from all patients, and it obtained an average sensitivity of 69% and an average alert time of 27%.

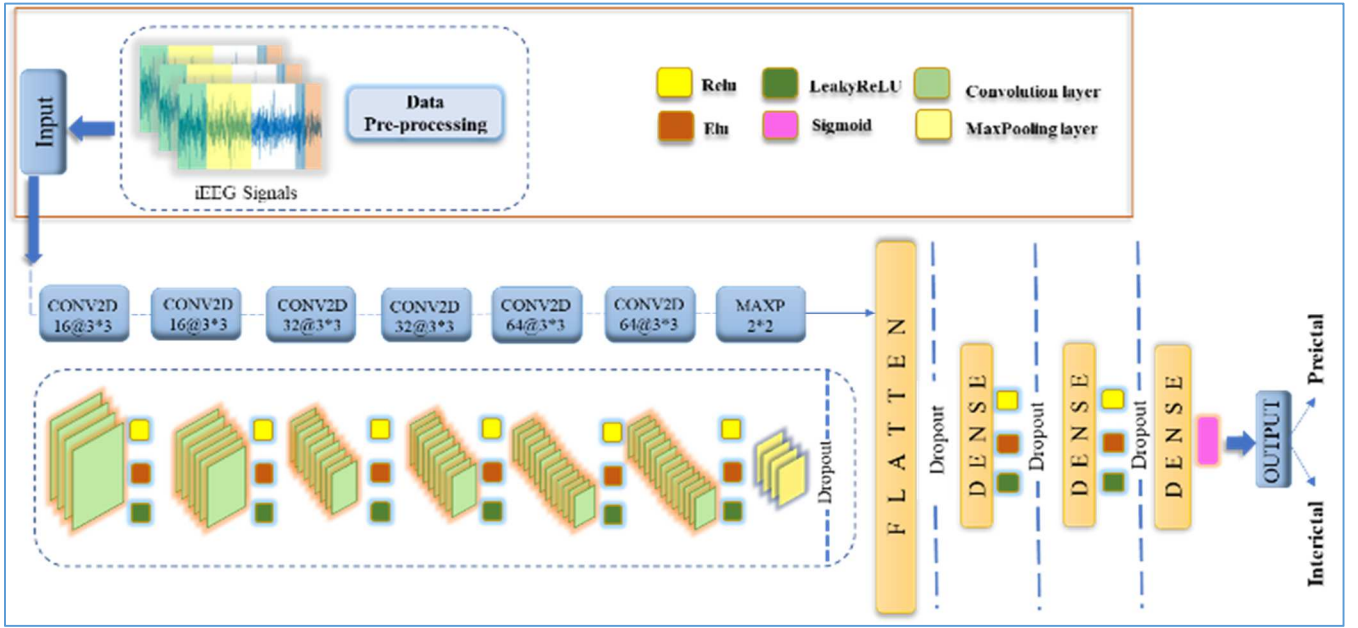


Fig. 1. CNN model.

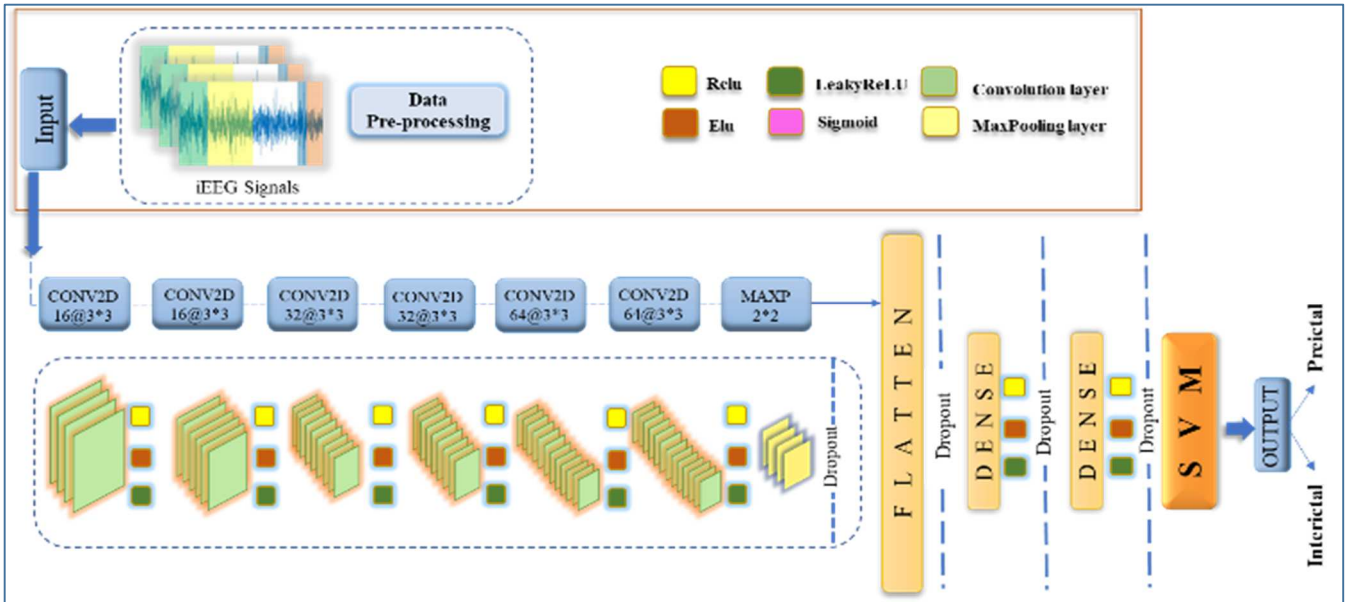


Fig. 2. CNN-SVM model.

In [5] three different CNN topology models were tested on an iEEG dataset of seven different subjects. The prediction accuracy obtained for each subject is presented for each of the three investigated topologies. Furthermore, topology 1 outperforms the other models on the three patients in the iEEG human dataset (patient 1: 24.4%, patient 2: 73.7%, and patient 3: 72.1%).

The authors in [6] suggested a multi-view CNN framework for seizure prediction that takes into account time and frequency domains as inputs. In terms of the private score for the iEEG dataset, this model obtains an average area under the curve (AUC) of 0.84.

Deep Convolutional Generative Adversarial Network was proposed in [7] as an unsupervised feature extraction technique (DCGAN). The main idea is that Discriminator (D) and Generator (G) perform until they reach an equilibrium. This network's prediction of epileptic seizures has an AUC of 75.47%.

In [8] the authors proposed a method of combining frequency-domain and deep learning by using the DTF algorithm to extract channel-frequency maps and use them as input for the CNN model to predict seizures. Achieving an average prediction of sensitivity of 90.8% and a false prediction rate of 0.08 per hour.

To better exploit the geometry of rectangular images with a high aspect ratio, the authors in [9] proposed a neural network architecture based on a new convolution strategy known as "semi-dilated convolution". Based on iEEG data, this method has an average specificity and classification accuracy of 88.24% and 88.38%, respectively.

III. PROPOSED METHOD

A. Dataset description

In this study, we used the Epilepsy Seizure Prediction Challenge database from the American Epilepsy Society. Over 627 hours and 48 seizures, it contains intracranial EEG

readings from five dogs and two patients. Patients 1 and 2 had their iEEG data recorded at a rate of 5 kHz with 15 deep electrodes and 24 subdural electrodes, respectively. Each seizure has six pre-ictal segments available up to five minutes before the start of the crisis, and this database contains 10-minute segments of preictal and interictal data[10].

To keep things simple and avoid running out of drive space, we trained the models on data from a single patient, where each channel is also representative of the characteristics learned by our model to determine whether or not the segment precedes a seizure.

B. Pre-processing

In this study, we used a simple preprocessing method that consists of transforming the iEEG recordings into two-dimensional plots, with all frequencies displayed in the height and all-time steps presented in the width. The matrices are resized to 256×22 for classification. We gathered 21600 samples and split them into 90% for training and 10% for testing our models after shuffling the data.

C. The proposed models

This paper proposes a classification method based on deep convolutional neural networks to classify two critical epileptic states using iEEG signals. The three major components of the models depicted in figures 1 and 2 are iEEG data preprocessing, feature extraction, and classification. Two methods for identifying and classifying preictal and interictal iEEG signals were proposed:

a) Seizure prediction using CNN

Figure 1 depicts our first model, a simple CNN model. The feature extraction step employs six convolutional layers. The first two convolutional layers each have 16 filters with an activation function, the second two have 32 filters, and the last two have 64 filters with a kernel size of 3×3. A max-pooling layer is used after the last convolution layer, followed by a dropout with a probability of 0.5.

The classification step then begins with a flattened layer, which converts the input into a one-dimensional list for input into the dense layers. Finally, three dense layers are chained together with a 0.5 dropout between them.

b) Seizure prediction using CNN-SVM

A merge between deep learning and machine learning is used in this architecture. Based on the first model (CNN), we created another model called CNN-SVM, which is depicted in Figure 2. It employs an SVM classification method, which replaces the last fully connected layer (i.e., the activation of the last dense layer serves as an input to the SVM) and is trained using automatically derived feature vectors.

SVM is a supervised machine learning technique used to address regression and classification issues. It is based on statistical learning theory and the structural risk minimization concept. It is capable of solving both linear and nonlinear problems. Because we have a two-class classification issue, we employ a linear SVM.

IV. EXPERIMENTAL RESULTS

We present the binary, preictal, and interictal iEEG signal categorization findings of our CNN and CNN-SVM models in this section. We ran the models through 50, 100, and 150

epochs of optimization. The batch size is set to 128, and the loss function is optimized using the RMSprop optimizer. The CNN model is trained using a binary cross-entropy loss function, while the CNN-SVM model uses a hinge loss function. Relu, Elu, and LeakyReLU were used and evaluated as the activation functions in these two models.

We created the models in Google Colaboratory using Python 3 and the Keras framework, which runs on the TensorFlow platform.

A. Evaluation metrics

A confusion matrix is a table that is used to define a classification algorithm's performance. It can be used for binary categorization, as in our study. It consists of four main properties that establish the classifier's measuring metrics. As shown in Table 1, the confusion matrix for two classes of classifiers is as follows:

Table 1. Confusion matrix for binary classification.

	Positive (Actual)	Negative (Actual)
Positive (Predicted)	TP	FN
Negative (Predicted)	FP	TN

True positive (TP): correctly predicts the positive class.

True negative (TN): correctly predicts the negative class.

False-positive (FP): incorrectly predicts the negative class as positive.

False-negative (FN): incorrectly predicts the positive class as negative.

We assessed our methods using a variety of confusion matrix indicators such as Accuracy, Precision, Recall, and F1 score. They can be computed using the following formula:

$$\text{Accuracy} = \frac{(TP+TN)}{(TP+TN+FP+FN)} \quad (1)$$

$$\text{Precision} = \frac{TP}{FP+TP} \quad (2)$$

$$\text{Recall} = \frac{TP}{FN+TP} \quad (3)$$

$$\text{F1-score} = 2 \frac{(\text{Recall} \times \text{Precision})}{(\text{Recall} + \text{Precision})} \quad (4)$$

Accuracy can be defined as the ratio of correctly predicted observations to total observations. Precision refers to how precise the model is in terms of how many of those predicted positives are positive. Recall computes how many actual positives our model captures by labeling it as positive. The harmonic mean of precision and recall is the F1 score.

Furthermore, the confusion matrix provides a methodical technique for illustrating the classification accuracy for the two groups of epileptic states in this study. The true positive rate (TP) refers to the number of segments correctly classified as Interictal EEG, whereas the true negative rate (TN) refers to the number of segments correctly classified as Preictal EEG.

B. Results and Discussion

In this research, two deep learning models are proposed that use one patient's iEEG signals to distinguish two brain

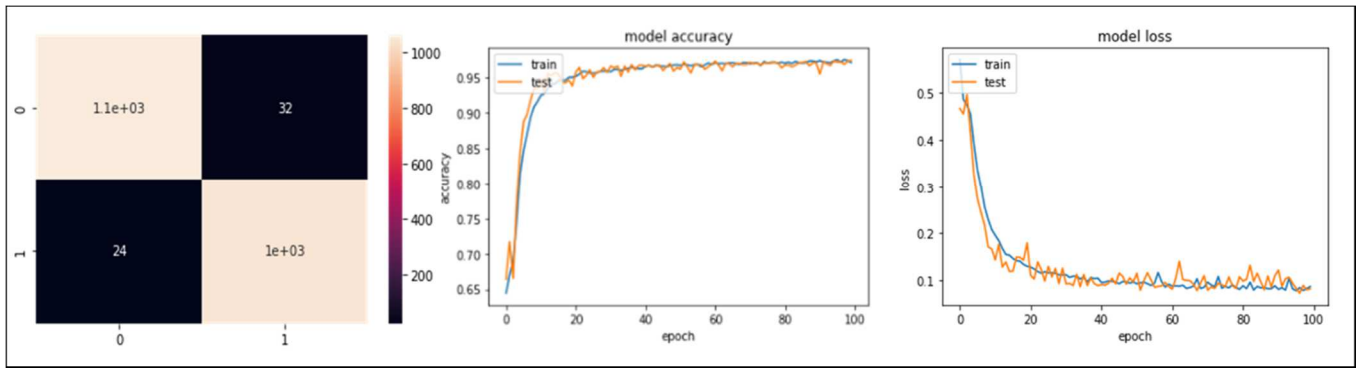


Fig. 3. The confusion matrix, Accuracy, and Loss plots were obtained by the CNN model with the Elu activation function after 100 epochs.

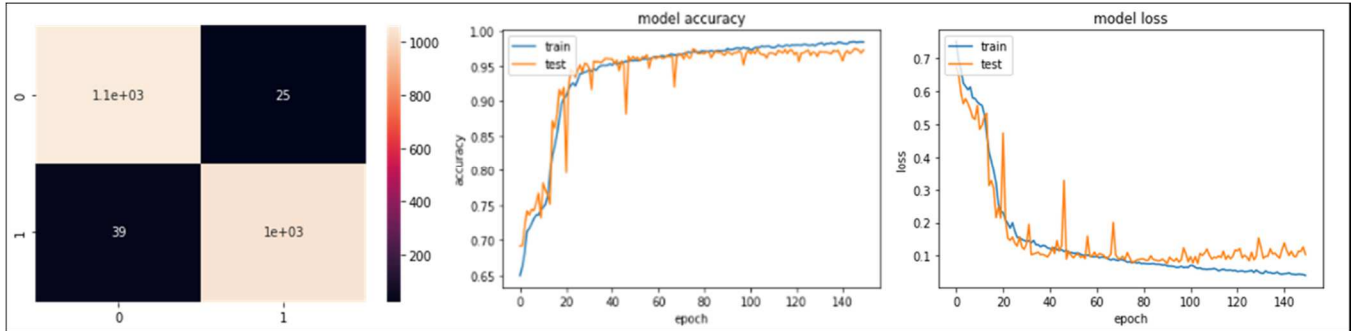


Fig. 4. The confusion matrix, Accuracy, and Loss plots were obtained by the CNN-SVM model with LeakyRelu activation function after 150 epochs.

states of the epileptic seizure (Interictal and Preictal), which may help in seizure prediction.

The classification reports of the CNN and CNN-SVM models used with the various activation functions are presented in tables 2 and 3, respectively. After 100 epochs, the model CNN with the activation functions Elu produces the best accuracy result of 97.41 percent. However, after 150 epochs, CNN-SVM achieved the best accuracy with the LeakyRelu activation function. Convolutional Neural Networks capture many high-level features from iEEG data, which is why CNN-SVM achieved the best training result of 98.45 percent.

Table 2. Accuracy value of the CNN model.

	ReLU		LeakyRelu		Elu	
	Train	Test	Train	Test	Train	Test
50 epochs	94.09%	93.98%	95.24%	95.15%	96.46%	96.85%
100 epochs	95.52%	96.20%	97.21%	96.48%	97.16%	97.41%
150 epochs	94.99%	73.94%	97.29%	96.67%	97.61%	96.71%

Table 3. Accuracy value of the CNN-SVM model.

	ReLU		LeakyRelu		Elu	
	Train	Test	Train	Test	Train	Test
50 epochs	96.85%	96.85%	95.80%	95.97%	96.51%	96.30%
100 epochs	97.17%	96.53%	97.41%	95.69%	97.68%	96.25%
150 epochs	98.27%	94.54%	98.45%	97.31%	97.18%	96.99%

In terms of classification percentage, we can see that the CNN is slightly better than the CNN-SVM. The reason for this is that the linear classification boundary in SVM requires the sample data in the testing set to be distributed consistently.

The best results, of Precision, Recall, and F1-score, were achieved with the activation function Elu for the CNN model and with LeakyRelu for the CNN-SVM model, as shown in Tables 4 and 5. It's worth noting that the CNN model outperforms the CNN-SVM model in terms of performance accuracy, with an average rate of 97.41%, and in terms of Precision and Recall, with an average rate of 97.5%. However, they are equal in terms of F1 score, with a 97% average rate. So, in terms of Precision and Recall, the CNN model surpasses the CNN-SVM model by 0.5%. This is a desirable outcome because a high Recall value leads to a low number of false negatives. These significant findings show that the number of false negatives is low, which is consistent with the study's main goal.

Table 4. Classification report of the CNN model.

	Precision	Recall	F1-score
Interictal	98%	97%	97%
Preictal	97%	98%	97%
Mean	97.5%	97.5%	97%

Table 5. Classification report of the CNN-SVM model.

	Precision	Recall	F1-score
Interictal	97%	97%	97%
Preictal	97%	97%	97%
Mean	97%	97%	97%

The loss and accuracy between the training and validation phases are depicted in figures 3 and 4 for the two models. Figure 3 shows the CNN model performance with the activation function Elu after 100 epochs, which is the model's best result. After each optimization iteration, the loss value indicates how well or poorly a model performs. The lower the loss, the better the model; as shown in figure 3, the CNN model achieves the best loss value of 0.0816.

In comparison to the other model (CNN-SVM), the CNN model correctly classified the interictal and preictal segments, achieving the best classification with the fewest false negatives and false positives (56 misclassified segments for CNN and 64 misclassified segments for CNN-SVM). This indicates that TP and TN have a high value (2104 correctly classified segments for CNN and 2096 correctly classified segments for CNN-SVM), and therefore this approach produces a high prediction percentage.

The experimental results show that the classification with the CNN model improves overall performance by around 1% compared to CNN-SVM. However, after 150 epochs with LeakyRelu, the classification with CNN-SVM improves accuracy by about 0.5 percent compared to the CNN classification. We can then conclude that the CNN model has higher overall accuracy as compared to the CNN-SVM.

Even though the proposed CNN model is suitable for the classification problem of the used iEEG dataset and achieved an accuracy of 97.41%, the experiment still needs to be carried out to increase the number of epochs, which will certainly improve the accuracy of seizure state recognition but will require more time for training. And because we used a data set of one patient, we can try to improve the model by increasing the data of other patients.

Table 6. A comparison of automatic seizure prediction methods and the methods we proposed.

References	Model	Accuracy
[11]	LSTM-MTL	92.37%
[12]	MAML: SVM, CNN, LSTM	94.96%
Proposed models	CNN-SVM	97.31%
	CNN	97.41%

Table 6 shows a summary of the results of the automatic prediction of epileptic seizures and a comparison to our suggested model. In terms of accuracy, the summary shows that our submitted model outperforms other models.

The authors in [11] suggested the use of LSTM-based multi-task learning for seizure prediction, extracting a set of time-domain features from iEEG data and feeding them into the LSTM-MTL framework, attaining an average accuracy score of 89.36% for the two humans (patient 1 and patient 2) and 92.37% for patient 1.

In [12], the authors employed a technique known as Few-Shot learning to develop a supervised machine learning approach. They propose a Model Agnostic Meta-Learning classifier that gets output from three distinct classifiers: SVM, CNN, and LSTM. This model has a sensitivity of 94.20%, a

specificity of 95.81%, an accuracy of 95.53% on average for all subjects, and an accuracy of 94.96% particularly for patient 1.

The visual and experimental results demonstrated the superior performance of our method in seizure prediction achieving a high accuracy score of 97.41%. Furthermore, when compared to existing studies shown in table 6, our proposed model shows a significant improvement in seizure prediction performance. Meaningfully, the proposed CNN model is remarkably successful, which can provide a promising solution for iEEG-based seizure prediction, making it suitable for use in clinical and medical applications.

V. CONCLUSION

This study focuses on predicting seizures of persistent epileptic conditions based on iEEG recordings using CNN, a deep neural network approach. This network is a component of a categorization scheme. We used our suggested models to distinguish two brain states associated with epilepsy, namely preictal and interictal states, to perform seizure prediction to improve seizure control. Therefore, we presented two based models: CNN and CNN-SVM. Our proposed method outperforms current methods by predicting seizures with an average accuracy of 97.41 percent for the CNN model and 97.31 percent for the CNN-SVM model. This study offered a promising solution for seizure prediction based on iEEG data.

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